

KUSHANK AGGARWAL

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EDUCATION

Delhi Technological University, Delhi

Bachelor of Technology, Mechanical Engineering with Specialization in Automotive Engineering

August 2023 – May 2027

CGPA: 8.41/10

EXPERIENCE

Mechanical Intern | Creatara Mobility Pvt. Ltd., New Delhi

June 2026 – July 2026

- Built a seat-occupancy sensing system that enables a vehicle USP by integrating rider detection into the control architecture, unlocking 5+ vehicle level features including occupant detection, overload protection and adaptive torque control, while identifying an 8-9% sensing variation caused by load-transfer geometry, driving the next design iteration towards production-level robustness.
- Proposed a mechanical redundancy mechanism for an electronic steering lock and implemented a steering and vehicular angle measurement system with 0.088° resolution, establishing a common platform capable of supporting future software-defined vehicle features such as steering lock validation, cornering ABS and yaw based stability control. · [Certificate](#)

Summer Research Intern | Indian Institute of Technology, Delhi

May 2025 – July 2025

- Developed a high precision displacement sensor by analyzing strain patterns and material deformation characteristics, increasing measurement sensitivity by 57.5% over a conventional design, validated through theoretical formulations and FEA within 1% deviation.
- Implemented MATLAB-based data visualization and calibrated a working prototype using half-bridge strain gauge circuit, achieving a linear response over a ±6 mm range with ±3µm accuracy against a Keyence GT2 reference sensor.
- First author, "Geometric Modification of a Cantilever Element for Enhanced Displacement Measurement Using Strain Gauges," submitted to IPRoMM 2026 (Springer Nature) — under review.

Mechanical Trainee | Mecrox Tech Pvt. Ltd., Gurugram

December 2024 – January 2025

- Facilitated in the prototype manufacturing of the brake pedal assembly for Surge 32, a convertible 2-in-1 electric vehicle project by Hero MotoCorp.
- Generated laser-cut manufacturing drawings with GD&T, maintaining dimensional tolerances up to ±0.02 mm.
- Executed hands-on fabrication processes including grinding, buffing, cutting, and welding, gaining 60+ hours of practical experience with industrial machinery. · [Certificate](#)

Mechanical Design Intern | Innovante Engineering Solutions, Gurugram

May 2024 – July 2024

- Designed a water-resistant PCB enclosure with a O-ring sealing system, achieving 20–25% controlled compression through tolerance stack-up aligned with IP67 design principles to ensure 1m immersion protection.
- Identified gaps through benchmarking 10+ tracking devices and proposed modular attachment extensions, expanding potential use cases to 3 distinct user segments.
- Translated conceptual requirements for Wearenote.com into 5+ manufacturable mechanical designs. · [Certificate](#)

SKILLS

CAD & Simulation: ANSYS Workbench (DesignModeler, Mechanical), SolidWorks (Weldments, Surfacing), Autodesk Fusion 360

Analysis & Programming: Finite Element Analysis (FEA), MATLAB, Python

Design & Manufacturing: GD&T, DFM, Composite Manufacturing (E-glass, epoxy), Rapid Prototyping, Fabrication (welding, grinding, metal forming)

PROJECTS

Arka | Bharat Solar Vehicle Challenge (BSVC), 2025

July 2025 – October 2025

- Manufactured an aerofoil-shaped solar vehicle chassis and body architecture, achieving a drag coefficient (Cd) of 0.13 through CFD-driven aerodynamic refinement and surface optimization.
- Engineered and validated a 32kg tubular space-frame using SolidWorks Weldments, delivering torsional rigidity of 1800 Nm/deg through FEA-based structural validation.
- Integrated AISI 4130 Chromoly steel members, leveraging high strength-to-weight ratio of around 0.06, fatigue resistance, and weld integrity to meet BSVC safety compliance.
- Developed and produced a lightweight composite body using E-glass fibre, core mat, and epoxy resin system through hand layup process, ensuring dimensional stability and aerodynamic contour finish.

POSITIONS OF RESPONSIBILITY

Head Aerodynamics and Chassis | Team DTU Supermileage

September 2023 – May 2026

- Supervised a 7-member sub-team within a 50+ member vehicle team, achieving a 35% reduction in drag coefficient (Cd) for Project Arka through iterative CFD validation.
- Modeled and supervised tubular chassis fabrication and composite body manufacturing, overseeing complete vehicle assembly, representing DTU at Bharat Solar Vehicle Challenge (BSVC) 2025, Coimbatore.

Vice President | American Society for Mechanical Engineers, DTU

September 2023 – May 2026

- Led a 30-member core team and 80+ volunteers, organizing 6+ annual technical seminars and competitions, including a flagship CAD workshop engaging more than 200 participants.
- Drove an 80% growth in active membership through engaging PR stunts and established a dedicated R&D division to promote participation in national-level competitions such as Smart India Hackathon.